Day 1 – Introduction to AI

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What is Artificial Intelligence?

Artificial Intelligence (AI) is a technology that enables machines and computers to perform tasks that normally require human intelligence.

Examples

- ChatGPT
- Face Recognition
- Google Maps
- Voice Assistants
- Recommendation Systems
- Self-Driving Cars

What is Machine Learning?

Machine Learning (ML) is a subset of AI where computers learn patterns from data and use those patterns to make predictions or decisions.

Example

An email system can learn from previous emails and identify whether a new email is spam or not.

What is Deep Learning?

Deep Learning is a subset of Machine Learning that uses multi-layered neural networks to learn complex patterns from large amounts of data.

Examples

- Image Recognition
- Speech Recognition
- Generative AI

AI vs ML vs DL

AI	ML	DL
Broad field	Subset of AI	Subset of ML
Makes machines intelligent	Learns from data	Uses deep neural networks
Includes ML and DL	Uses algorithms to learn	Handles complex patterns

Real-World Applications of AI

1. Healthcare – Disease prediction and medical image analysis
2. Education – Personalized learning
3. Banking – Fraud detection
4. Transportation – Navigation and autonomous vehicles
5. Entertainment – Movie and music recommendations

Key Learning

I learned the basic concepts of AI, Machine Learning, and Deep Learning and understood their differences and real-world applications.

Day 2 – Python Basics

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Introduction to Python

Python is a high-level, easy-to-learn programming language widely used in AI, Machine Learning, Data Science, and Web Development.

Variables

Variables are used to store data.

Example:

```
name = "Arnesh"
age = 19
mark = 85.5
```

Basic Data Types

- int – Integer values
- float – Decimal values
- str – Text
- bool – True or False

Input and Output

```
name = input("Enter your name: ")
print("Hello", name)
```

Conditional Statements

Python uses if, elif, and else to make decisions.

```
num = int(input("Enter a number: "))

if num % 2 == 0:
    print("Even")
```

```
else:
    print("Odd")
```

Programs Practiced

- Addition of two numbers
- Even or Odd
- Largest of two numbers
- Positive or Negative

Key Learning

I learned Python variables, data types, input/output, operators, and conditional statements.

Day 3 – Loops and Lists

Day 3 – Loops and Lists

For Loop

A for loop is used to repeat a block of code for a specific number of times.

```
for i in range(1, 11):
    print(i)
```

While Loop

A while loop repeats the code as long as the condition is true.

```
i = 1

while i <= 5:
    print(i)
    i += 1
```

Lists

A list is used to store multiple values in a single variable.

```
numbers = [10, 20, 30, 40, 50]
```

Accessing List Elements

```
print(numbers[0])
```

Loop Through a List

```
for x in numbers:
    print(x)
```

Programs Practiced

- Print numbers from 1 to 100
- Print even numbers
- Print odd numbers
- Find sum of numbers
- Find largest number
- Find smallest number
- Find sum of list elements

Key Learning

I learned how to use for loops, while loops, and lists in Python.

Day 4 – Functions

Day 4 – Functions

What is a Function?

A function is a reusable block of code that performs a specific task.

Basic Function

```
def greet():
    print("Hello")

greet()
```

Function with Parameters

```
def add(a, b):
    return a + b

result = add(10, 20)
print(result)
```

Programs Practiced

Prime Number

Checked whether a number is prime or not.

Factorial

Calculated the factorial of a number using a loop.

Fibonacci

Generated Fibonacci numbers using a loop.

Key Learning

I learned how to create functions, pass parameters, return values, and write simple Python programs.

Module 1 Summary

During Day 1 to Day 4, I learned:

- Fundamentals of Artificial Intelligence
- Machine Learning and Deep Learning
- Real-world AI applications
- Python basics
- Variables and data types
- Conditional statements
- Loops
- Lists
- Functions
- Beginner Python programs

This module provided me with a basic foundation for learning Artificial Intelligence and Machine Learning.